

Data Mining

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Overview of topics

① Part I : Unsupervised-learning

- Introduction
- Clustering
- Applications with [Python](#)

② Part II : Supervised learning

- Linear, Quadratic, KNN, Discriminant Analysis,
- Classification : Logistic regression
- Classification : Regression trees, SVM, Bagging, Boosting,...
- Applications with [Python](#)

Some references

- Larry Wasserman (2004). All of statistics. A concise course in statistical inference. Springer texts in Statistics.
- Hastie, Tibshirani and Friedman (2009). The elements of Statistical Learning. Second edition.

Regression model

- Let (X, Y) a pair of real-valued variables such that Y is continuous X is p -dimensional, $p \geq 1$.
- We are interested in knowing how the values taken by Y differ from those taken by X .
- A model of regression consists in assuming that there exists a (measurable) function $f : \mathbb{R}^p \rightarrow \mathbb{R}$ such that

$$Y = f(X) + \varepsilon$$

- we have to find a function $r(\cdot)$ such that $r(X)$ is a “best approximation of Y ”.
- Then $|Y - r(X)|$ should be small.
- Because of the randomness of this deviation, we can minimize the quadratic risk function : $\mathbb{E}|Y - r(X)|^2$.
- Other risk functions are possible

- Find a function $r : \mathbb{R}^p \rightarrow \mathbb{R}$:

$$\mathbb{E}[Y - r(X)]^2 = \min_{f: \mathbb{R}^p \rightarrow \mathbb{R}} \mathbb{E}[Y - f(X)]^2.$$

- Since :

$$\mathbb{E}[Y - f(X)]^2 = \mathbb{E}[Y - \mathbb{E}(Y|X)]^2 + \mathbb{E}[f(X) - \mathbb{E}(Y|X)]^2$$

Then :

$$r(\cdot) = \mathbb{E}(Y|X = \cdot)$$

- In practice the probability distribution of $Y|X$ is unknown.

AIM :

Estimate the function $r(\cdot)$ with the sample $(X_1, Y_1), \dots, (X_n, Y_n)$.

- Linear regression (parametric)
- Non-linear regression (non-parametric)

Several regression methods/algorithms

- **Linear regression** : It aims to find a linear combination of features that best fits Y and X
- **KNN (K nearest neighbors)** : uses K nearest neighbors to create a weighted mean regression estimate
- **Decision Tree (DT)** : creates a tree-like structure based on informative features and make predictions by traversing the tree. DT is known for its simplicity and ability to capture non-linear relationships in data.
- **Generalized Linear Models (GLMs)** : GLMs are versatile models used for both regression and classification tasks. They allow a wide range of response variable distributions and relationships between predictors and the response. GLMs models are useful when the assumptions of traditional linear models do not hold.
- ...

The aim here is about predicting the unknown nature of an object, a discrete quantity for example one or zero.

- Let an observation of an object be a d -dimensional vector X . The unknown nature of the object is called a *class* and is denoted by Y which takes values in a finite set $\{1, \dots, M\}$.
- In classification, one constructs a function (*classifier*) g taking values in $\{1, \dots, M\}$ which represents one's guess $g(X)$ of Y given X .

The aim to predict the class Y from X using a sample of this pair of variables at some observations.

- A mapping g is then defined on $\mathbb{R}^d$ and takes values in $\{1, \dots, M\}$.
- One makes a mistake on Y if $g(X) \neq Y$, and the probability of error for a classifier g is given by :

$$L(g) = P\{g(X) \neq Y\}.$$

- It is well known that the Bayes classifier defined by,

$$g^* = \arg \min_{g: \mathcal{X} \rightarrow \{1, \dots, M\}} P\{g(X) \neq Y\},$$

is the best possible classifier with respect to the quadratic loss.

- The minimum probability of error is called the *Bayes error* and is denoted by $L^* = L(g^*)$. Note that g^* depends on the distribution of (X, Y) which is unknown.

An estimator g_n of g is based on the observations $((X_i, Y_i)_{i=1, \dots, n})$; Y is predicted by $g_n(X; (X_i, Y_i)_{i=1, \dots, n})$. The performance of g_n is measured by the conditional probability of error

$$L_n = L(g_n) = P\{g_n(X; (X_i, Y_i)_{i=1, \dots, n}) \neq Y\} \geq L^*.$$

The sequence $\{g_n, \quad n \in \mathbb{N}\}$ is the *discrimination rule*.

Several classification methods/algorithms

- **KNN (K nearest neighbors classifier)** : uses K nearest neighbors to create a weighted classification rule
- **Linear Discriminant Analysis (LDA)** : is a method for both dimensionality reduction and classification tasks. It aims to find a linear combination of features that maximizes the separation between different classes. It assumes equal covariance matrices for all classes.
- **Quadratic Discriminant Analysis (QDA)** : is a classification algorithm that allows for the creation of non-linear decision boundaries, it is useful for dealing with complex data distributions or when the assumptions of LDA are not met.
- **Decision Tree Classification (DT)** : creates a tree-like structure to partition data based on informative features and make predictions by traversing the tree. DT is known for its simplicity and ability to capture non-linear relationships in data.
- **Generalized Linear Models (GLMs)** : GLMs are versatile models used for both regression and classification tasks. They allow a wide range of response variable distributions and relationships between predictors and the response. Logistic model is a particularly case.